Code No: 117AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, May/June - 2019
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is Artificial Intelligence? [2]
b) Define state space search. [3]
c) What are the basic components in a semantic network? [2]
d) Represent the following sentences in predicate logic:
i) Paris and Marseilles are both in France.
ii) There is a country that borders both Iraq and Pakistan.
iii) All countries that border Ecuador are in South America. [3]
e) Give an application of expert system in medicine. [2]
f) Give Baye's rule and its uses. [3]
g) What do you mean by supervised learning? Give examples of supervised learning. [2]
h) Give the schematics of a single layer perceptron. [3]
i) Give the applications of natural language processing. [2]
j) What is a syntax tree? [3]

PART-B

(50 Marks)

2. Demonstrate Depth-First Iterative Deepening (DFID) algorithm with an example and list the advantages and disadvantages of DFID over depth-first search algorithm. [10]

OR

3.a) What do you mean by a control strategy? Give an overview of various control strategies for search.
b) Explain bounded-look ahead searching with an example. [5+5]

4.a) Give an overview of various knowledge representation approaches.
b) Represent the following knowledge using a semantic network: [5+5]
Tom is a cat
Tom caught a bird
Tom is owned by John
Tom is ginger in colour
Cats like cream
Cat is on the mat
A cat is a mammal
A bird is an animal
All mammals are animals
Mammals have fur

OR

JNTUH USED 10-06-2019 PM

5.a) Discuss how inference through inheritance can be achieved in a frame based knowledge representation.
b) Give the rules of inference in propositional logic and explain natural deduction using an example. [5+5]

6.a) Give an overview of knowledge base and inference engine in expert systems.
b) Distinguish among the following Truth Maintenance Systems: JTMS, ATMS and LTMS. [5+5]

OR

7.a) Give an overview of various types of expert system tools and criteria for selecting the right kind of tool.
b) Give an overview of reasoning with uncertainty using Dempster-Shafer theory. [5+5]

8. Explain how entropy and information gain are used in the construction of decision tree. [10]

OR

9. Give an overview of multi-layer perceptron algorithm in detail. [10]

10.a) What are the tasks involved in natural language processing? Explain.
b) What is the purpose of a case grammar? Give an overview of knowledge representation using case grammars with examples. [5+5]

OR

11.a) Describe Universal Networking Language for representing semantic knowledge.
b) What is a semantic web? Give an example of a semantic web and list the applications of semantic web. [5+5]

--ooOoo--

JNTUHUSED 10-06-2019PM